



LINEAR ALGEBRA I

Chapter 2. Euclidean Vector Spaces

Boqing Xue



WELCOME!

2023 Fall



Chapter 2. Euclidean Vector Spaces

§2.1 Basics of Vectors

§2.2 Dot Product and Orthogonality

§2.3 Cross Product and Scalar Triple Product

§2.4 General Determinants

§2.5 Properties of Determinants; Cramer's Rule

§2.6 Matrix Transformations



Representations of Vectors

- Geometric vector: an arrow in n -dimensional space (called n -space).
(e.g., $\mathbf{v}$ or $\overrightarrow{PQ}$, where P is initial point and Q is terminal point, $\mathbf{u} = \mathbf{v}$.)



- n -space $\mathbb{R}^n = \{\mathbf{v} = (v_1, v_2, \dots, v_n) : v_1, \dots, v_n \in \mathbb{R}\}$.
- Vector in a Coordinate System: $\mathbf{v} = (v_1, v_2, \dots, v_n)$.
- Column-matrix form of a vector:

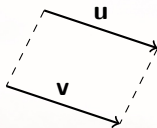
$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

- zero vector $\mathbf{0}$: having length 0, and arbitrary direction.
(e.g., $\overrightarrow{PP}$, $\mathbf{0} = (0, 0, \dots, 0) \in \mathbb{R}^n$, $\mathbf{0} \in M_{n \times 1}$.)



Equality of Vectors

- **Equal** vectors: having same direction and length. (e.g., $\mathbf{u} = \mathbf{v}$.)



- Equality of vectors:

$$(u_1, u_2, \dots, u_n) = (v_1, v_2, \dots, v_n) \iff u_1 = v_1, \dots, u_n = v_n.$$

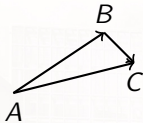
- Equality of vectors:

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \iff u_1 = v_1, \dots, u_n = v_n.$$



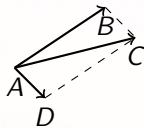
Addition

- Sum: $\overrightarrow{AB} + \overrightarrow{BC} \stackrel{\text{def}}{=} \overrightarrow{AC}$

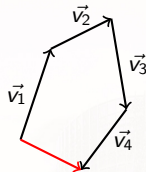


Triangle Rule

$$\overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AC}$$



Parallelogram Rule



$\vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \vec{v}_4$

- $(u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n) = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n).$

- $$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

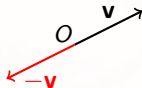
- Associative law for addition: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w}).$



Difference of Vectors

- The negative of a vector:

$$\overrightarrow{PQ} + \overrightarrow{QP} = \overrightarrow{PP} = \vec{0} \implies \text{denoted by } \overrightarrow{QP} = -\overrightarrow{PQ}.$$



- The difference of vectors:

$$\mathbf{a} - \mathbf{b} \stackrel{\text{def}}{=} \mathbf{a} + (-\mathbf{b}),$$

$$\overrightarrow{OA} - \overrightarrow{OB} = \overrightarrow{BA}.$$

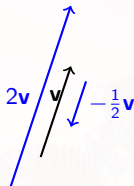


Scalar Product

- Scalar Products

The scalar product $k\mathbf{v}$ of a vector $\mathbf{v}$ by a scalar k :

- ◇ $\|k\mathbf{v}\| = |k| \cdot \|\mathbf{v}\|$.
- ◇ The direction of $k\mathbf{v}$ is
 - same as that of $\mathbf{v}$ when $k > 0$;
 - opposite to that of $\mathbf{v}$ when $k < 0$;
 - equal to $\mathbf{0}$ when $k = 0$.



- $k(v_1, v_2, \dots, v_n) = (kv_1, kv_2, \dots, kv_n).$

-

$$c \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} cu_1 \\ cu_2 \\ \vdots \\ cu_n \end{bmatrix}$$



Basic Properties of Vector Operations

Theorem. Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and λ, μ be scalars.

- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}.$
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}.$
- $\mathbf{v} + \vec{0} = \mathbf{v}.$
- $\mathbf{v} + (-\mathbf{v}) = \vec{0}.$
- $\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v}.$
- $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}.$
- $\mu(\lambda\mathbf{v}) = (\mu\lambda)\mathbf{v}.$
- $1\mathbf{v} = \mathbf{v}.$
- $0\mathbf{v} = \vec{0}.$
- $\lambda\vec{0} = \vec{0}.$
- $(-1)\mathbf{v} = -\mathbf{v}.$



Linear Combinations

Definition. Let $\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ be vectors in $\mathbb{R}^n$. The vector $\mathbf{w}$ is said to be a **linear combination** of $\{\mathbf{v}_i\}_{i=1}^r$ if it can be expressed in the form

$$\mathbf{w} = k_1 \mathbf{v}_1 + k_2 \mathbf{v}_2 + \dots + k_r \mathbf{v}_r$$

where the scalars $k_1, k_2, \dots, k_r$ are called **coefficients** of the linear combination.

- In column-matrix form, a vector $\mathbf{w}$ is a linear combination of $\{\mathbf{v}_i\}_{i=1}^r$ if and only if

$$\begin{bmatrix} \mathbf{w} \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_r \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_r \end{bmatrix}.$$

for some scalars $k_1, k_2, \dots, k_r$.



An Example

Linear system:

$$\begin{cases} x + 3y = 2, \\ 2x - y = 3. \end{cases}$$

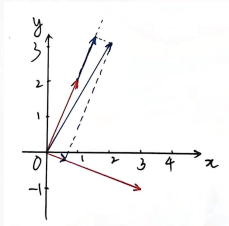
Matrix form:

$$\left[\begin{array}{c|c} 1 & 3 \\ 2 & -1 \end{array} \right] \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Linear combination:

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Figure in xOy :





An Example

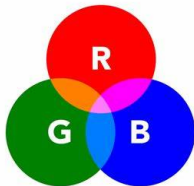
Example. Let $\mathbf{u} = (2, 1, 0, 1)$ and $\mathbf{v} = (-2, 3, 1, 0)$. Is $\mathbf{w} = (-8, 8, 3, -1)$ a linear combination of $\mathbf{u}$ and $\mathbf{v}$?

Solution:

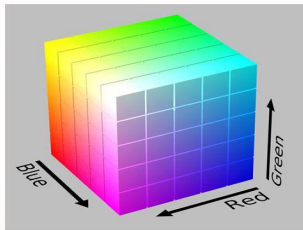
①



RGB Color

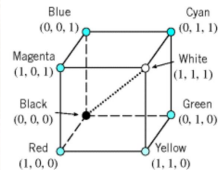


h



- Pioneers of RGB:
Thomas Young, James Maxwell and Hermann Helmholtz.
- Principle: superposition of waves/Fourier analysis.

$r = (1, 0, 0)$ (pure red),
 $g = (0, 1, 0)$ (pure green),
 $b = (0, 0, 1)$ (pure blue)





Norm of a Vector

Definition. Let $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be a vector in $\mathbb{R}^n$, then the **norm** of $\mathbf{v}$ is defined by

$$\|\mathbf{v}\| = (v_1^2 + v_2^2 + \dots + v_n^2)^{1/2}.$$

- In geometric form, norm is the length of the arrow.
(e.g., $\|\mathbf{v}\|$, $\|\overrightarrow{PQ}\|$.)

Theorem. Let $\mathbf{v}$ be a vector in $\mathbb{R}^n$ and k be a scalar.

- ◇ $\|\mathbf{v}\| \geq 0$
- ◇ $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$
- ◇ $\|k\mathbf{v}\| = |k| \cdot \|\mathbf{v}\|$.



Unit Vector

- **Unit** vector: a vector with norm 1. (It determines a direction.)
- For non-zero vector $\mathbf{v}$,

$$\mathbf{u} = \|\mathbf{v}\| \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Here the vector $\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$ is called the **normalization** of $\mathbf{v}$.

Example. Find the unit vector $\mathbf{u}$ that has the same direction as $\mathbf{v} = (2, -1, -2)$.

Solution:

②



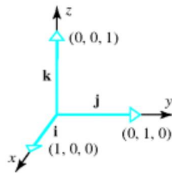
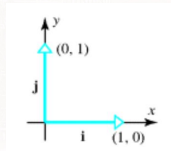
The Standard Unit Vectors

- In $\mathbb{R}^n$, the **standard unit vectors** are

$$\mathbf{e}_1 = (1, 0, 0, \dots, 0), \mathbf{e}_2 = (0, 1, 0, \dots, 0), \dots, \mathbf{e}_n = (0, 0, 0, \dots, 1).$$

- In $\mathbb{R}^3$, we usually write

$$\mathbf{i} = (1, 0, 0), \quad \mathbf{j} = (0, 1, 0), \quad \mathbf{k} = (0, 0, 1).$$



- If $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$, then

$$\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \dots + v_n \mathbf{e}_n.$$



Distance between Vectors

- Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be points in $\mathbb{R}^n$. The **distance** between $\mathbf{u}$ and $\mathbf{v}$ is defined as

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| \\ &= ((u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2)^{1/2}. \end{aligned}$$

Example. Let $\mathbf{u} = (1, 3, -2, 7)$ and $\mathbf{v} = (0, 7, 2, 2)$. Find the distance $d(\mathbf{u}, \mathbf{v})$.

Solution:

③



Chapter 2. Euclidean Vector Spaces

§2.1 Basics of Vectors

§2.2 Dot Product and Orthogonality

§2.3 Cross Product and Scalar Triple Product

§2.4 General Determinants

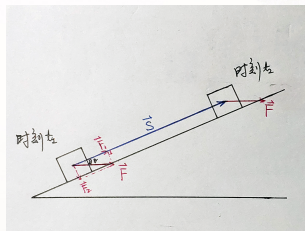
§2.5 Properties of Determinants; Cramer's Rule

§2.6 Matrix Transformations



Dot Product of Vectors (Geometric Form)

Problem: An object moves on the slope under a constant force $\mathbf{F}$ in the horizontal direction. And the displacement is $\mathbf{s}$. How much is the work done by $\mathbf{F}$?



Definition. If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$, and if θ is the angle between $\mathbf{u}$ and $\mathbf{v}$, then the **dot product** or **inner product** of $\mathbf{u}$ and $\mathbf{v}$ is defined as

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

If $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$, then we define $\mathbf{u} \cdot \mathbf{v} = 0$.

Remark: The dot product of two vectors is a scalar!



Dot Product of Vectors (Coordinative Form)

Definition. Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be vectors in $\mathbb{R}^n$. Then the **dot product** or **inner product** of $\mathbf{u}$ and $\mathbf{v}$ is defined as

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n.$$

Remark: In $\mathbb{R}^2$ or $\mathbb{R}^3$, the two definitions are related by “The Law of Cosines”.

Remark: Inner product contains information of “angle”.

Proposition. Let $\mathbf{u}, \mathbf{v}$ be vectors. Then

- $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$.
- $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$. (Cauchy's Inequality)

Corollary. Let $\mathbf{u}, \mathbf{v}$ be vectors in $\mathbb{R}^n$. We have

- ◊ $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$. (Triangular inequality)
- ◊ $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$. (Triangular inequality)

Remark: We will show proofs in the next chapter.



Properties of Inner Product

Proposition. Let k be a scalar and $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors. Then

- $\mathbf{v} \cdot \mathbf{u} = \mathbf{u} \cdot \mathbf{v}$. (Symmetry Property)
- $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$. (Homogeneity Property)
- $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$. (Distributive Property)

Example. Expand $(\mathbf{u} - 2\mathbf{v}) \cdot (3\mathbf{u} + 4\mathbf{v})$.

Solution:

④

Proposition. (Parallelogram Equation) Let $\mathbf{u}, \mathbf{v}$ be vectors in $\mathbb{R}^n$. Then

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$$

Proposition. (Polarization identity) Let $\mathbf{u}, \mathbf{v}$ be vectors in $\mathbb{R}^n$. Then

$$\mathbf{u} \cdot \mathbf{v} = \frac{1}{4} (\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2).$$

Remark: To prove something involving inner product, it is sufficient to prove it involving a norm.



Dot Product of Vectors (Column-matrix Form)

Definition. Let $\mathbf{u}, \mathbf{v} \in M_{n \times 1}$ be column vectors. Then the **dot product** or **inner product** of $\mathbf{u}$ and $\mathbf{v}$ is defined as

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v}^T \mathbf{u}.$$

Example. Suppose that $A \in M_{m \times n}$, $\mathbf{u} \in M_{n \times 1}$ and $\mathbf{v} \in M_{m \times 1}$. Verify that

$$A\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot A^T \mathbf{v}.$$

Solution:

⑤



Definition of Orthogonality

- Recall: Two vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ are said to be “perpendicular” if the angle between them is $\pi/2$.

Definition. $\diamond$ Two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$ are said to be **orthogonal** (or **perpendicular**) if $\mathbf{u} \cdot \mathbf{v} = 0$.

$\diamond$ A nonempty set of vectors in $\mathbb{R}^n$ is called an **orthogonal set** if all pairs of distinct vectors in the set are orthogonal.

$\diamond$ An orthogonal set of unit vectors is called an **orthonormal set**.

Example. (a) Show that $\mathbf{u} = (-2, 3, 1, 4)$ and $\mathbf{v} = (1, 2, 0, -1)$ are orthogonal vectors in $\mathbb{R}^4$.

(b) Show that the set $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is an orthonormal set in $\mathbb{R}^3$.

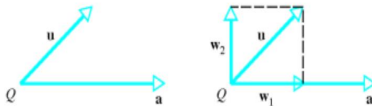
Solution:

⑥

Remark: An orthonormal set provides rectangular coordinate axes.



Projection Theorem



Theorem. Let $\mathbf{u}$ and $\mathbf{a}$ be vectors in $\mathbb{R}^n$ and $\mathbf{a} \neq 0$. Then $\mathbf{u}$ can be uniquely expressed as $\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_2$, where

- $\mathbf{w}_1$ is a scalar multiple of $\mathbf{a}$;
- $\mathbf{w}_2$ is orthogonal to $\mathbf{a}$.

Proof

⑦

- We denote $\text{proj}_{\mathbf{a}} \mathbf{u} = \mathbf{w}_1$, and call it the **orthogonal projection** of $\mathbf{u}$ on $\mathbf{a}$, or the **vector component** of $\mathbf{u}$ **along** $\mathbf{a}$. Indeed, $\text{proj}_{\mathbf{a}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2} \mathbf{a}$.
- We call $\mathbf{w}_2$ the **vector component** of $\mathbf{u}$ **orthogonal to** $\mathbf{a}$. Indeed, $\mathbf{w}_2 = \mathbf{u} - \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2} \mathbf{a}$.



Find the Projections

Example. Let $\mathbf{u} = (2, -1, 3)$ and $\mathbf{a} = (4, -1, 2)$. Find the vector component of $\mathbf{u}$ along $\mathbf{a}$ and the vector component of $\mathbf{u}$ orthogonal to $\mathbf{a}$.

Solution:

⑧

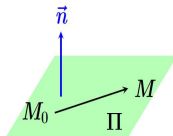
Theorem. (Gou-Gu Theorem) Suppose $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors in $\mathbb{R}^n$ with the Euclidean inner product. Then

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$



Planes in $\mathbb{R}^3$

- A non-zero vector $\mathbf{n}$ perpendicular to a plane is called a **normal**.



Proposition. Suppose that a plane through the point $M_0(x_0, y_0, z_0)$ has a normal $\mathbf{n} = (A, B, C)$. Then its equation, called **point-normal equation**, is given by

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

Proof

⑨

Proposition. Let A, B, C, D be constants such that A, B, C are not all zero. Then $Ax + By + Cz + D = 0$, called **normal form equation**, represents a plane in $\mathbb{R}^3$ with normal $\mathbf{n} = (A, B, C)$.

Question: What happens if (1) $D=0$; (2) $A=0$; (3) $A=B=0$.

Problem*: Try to determine lines in $\mathbb{R}^2$.



Lines in $\mathbb{R}^3$

Proposition. Suppose that a line through the point $M_0(x_0, y_0, z_0)$ is parallel to a non-zero vector $\mathbf{s} = (m, n, p)$. Then its equation, is given by

(i) **parametric form** $\mathbf{x} = \mathbf{x}_0 + t\mathbf{s}$ (here $\mathbf{x} = \overrightarrow{OM}$, $\mathbf{x}_0 = \overrightarrow{OM_0}$), i.e.,

$$\begin{cases} x = x_0 + tm, \\ y = y_0 + tn, \\ z = z_0 + tp, \end{cases}, \quad (t \in \mathbb{R}).$$

(i) **symmetric form** $\frac{x - x_0}{m} = \frac{y - y_0}{n} = \frac{z - z_0}{p}$.

(Here a denominator is zero means that the corresponding numerator is also zero.)

Proposition. Suppose that two planes $\Pi_i : A_i x + B_i y + C_i z = D_i$ ($i = 1, 2$) in $\mathbb{R}^3$ are not parallel. Then their intersection

$$\begin{cases} A_1 x + B_1 y + C_1 z + D_1 = 0, \\ A_2 x + B_2 y + C_2 z + D_2 = 0. \end{cases}$$

represents a line in $\mathbb{R}^3$.

Remark: Not parallel $\Leftrightarrow A_i : B_i : C_i$ are distinct $\Leftrightarrow$ rank of $\begin{bmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \end{bmatrix}$ is 2.



Lines in $\mathbb{R}^3$

Example. Suppose that a line is given by the intersection of two planes that are not parallel and through the origin, i.e.,

$$\begin{cases} A_1x + B_1y + C_1z = 0, \\ A_2x + B_2y + C_2z = 0. \end{cases}$$

Find its symmetric form.

Solution:

10



Chapter 2. Euclidean Vector Spaces

§2.1 Basics of Vectors

§2.2 Dot Product and Orthogonality

§2.3 Cross Product and Scalar Triple Product

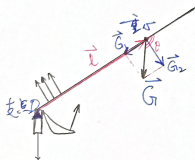
§2.4 General Determinants

§2.5 Properties of Determinants; Cramer's Rule

§2.6 Matrix Transformations



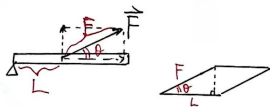
Background of Cross Product



- High-school-case (2d): The torque is defined by

$$F L \sin \theta.$$

Here F is the magnitude of the force,
 L is the length of the arm,
and θ is the angle between them.



Remark: It equals the **area** of corresponding parallelogram.

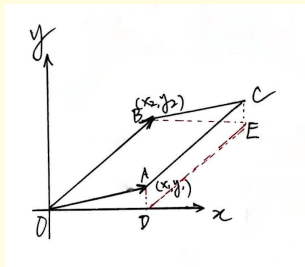


Geometric Meaning of a 2×2 determinant

Example. The vectors $\vec{OA} = (x_1, y_1)$ and $\vec{OB} = (x_2, y_2)$ are shown in the figure. Prove that the area of the parallelogram $OACB$ equals

$$\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} = x_1 y_2 - x_2 y_1.$$

Solution:

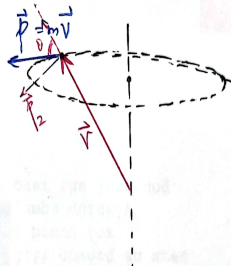
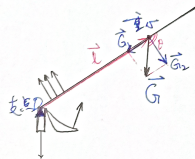


11

Remark: A 2×2 determinant represents the “signed area” of the parallelogram spanned by its row (or column) vector.



Background of Cross Product



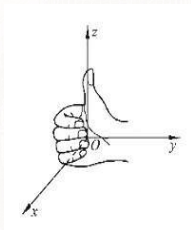
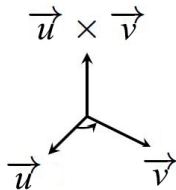
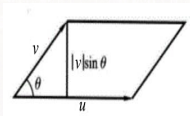


Cross Product of Vectors

Remark: In this section, all the vectors are in $\mathbb{R}^3$.

Definition. Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in $\mathbb{R}^3$ with angle θ . The **cross product** of $\mathbf{u}$ and $\mathbf{v}$, denoted by $\vec{u} \times \vec{v}$ is a vector in $\mathbb{R}^3$ satisfying the following conditions:

- $\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin \theta$
(the area of the parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$);
- $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$,
and its direction is determined by “right-hand rule”.



Remark: The cross product of two vectors is a vector.

Remark: $\vec{u} \times \vec{v} = \vec{0} \iff \vec{u}, \vec{v}$ are parallel.



Properties of Cross Products

Example. The standard unit vectors in $\mathbb{R}^3$ satisfy

$$\begin{aligned} \mathbf{i} \times \mathbf{j} &= \mathbf{k}, & \mathbf{j} \times \mathbf{k} &= \mathbf{i}, & \mathbf{k} \times \mathbf{i} &= \mathbf{j}, \\ \mathbf{j} \times \mathbf{i} &= -\mathbf{k}, & \mathbf{k} \times \mathbf{j} &= -\mathbf{i}, & \mathbf{i} \times \mathbf{k} &= -\mathbf{j}. \end{aligned}$$

Try to compute $\vec{i} \times (\vec{j} \times \vec{j})$ and $(\vec{i} \times \vec{j}) \times \vec{j}$.

Theorem. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors in $\mathbb{R}^3$ and k be a scalar.

- ◇ $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$.
- ◇ $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 0$.
- ◇ $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = 0$.
- ◇ $k(\mathbf{u} \times \mathbf{v}) = (k\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (k\mathbf{v})$.
- ◇ $\mathbf{u} \times \mathbf{0} = \mathbf{0} \times \mathbf{u} = \mathbf{0}$.
- ◇ $\mathbf{u} \times \mathbf{u} = \mathbf{0}$.
- ◇ $\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2$.

Proof:

12



Cross Product of Vectors

Theorem. Suppose that $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$. Then

$$\vec{u} \times \vec{v} = \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}, \begin{vmatrix} u_3 & u_1 \\ v_3 & v_1 \end{vmatrix}, \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \right).$$

Proof:

⑬

Example. Find a unit vector that is orthogonal to both $\vec{u} = (3, -2, 4)$ and $\vec{v} = (1, 1, -2)$.

Solution:

⑭

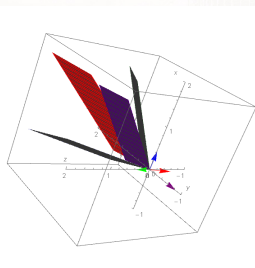
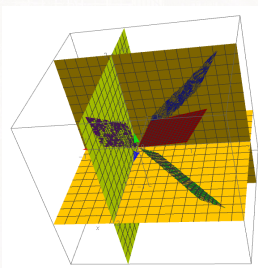


Gou-Gu Theorem for Signed Areas

Theorem. Suppose that $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$. Then

$$\begin{aligned}\|\mathbf{u} \times \mathbf{v}\|^2 &= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}^2 + \begin{vmatrix} u_3 & u_1 \\ v_3 & v_1 \end{vmatrix}^2 + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}^2 \\ &= \|\mathbf{u}_{yOz} \times \mathbf{v}_{yOz}\|^2 + \|\mathbf{u}_{xOz} \times \mathbf{v}_{xOz}\|^2 + \|\mathbf{u}_{xOy} \times \mathbf{v}_{xOy}\|^2.\end{aligned}$$

Here $\mathbf{u}_{yOz}$ is the projection of $\mathbf{u}$ on the yOz -plane, and the other notations are similar.





Properties involving Cross Products

Theorem. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors in $\mathbb{R}^3$.

$$\diamond \mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w}).$$

$$\diamond (\mathbf{u} + \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \times \mathbf{w}) + (\mathbf{v} \times \mathbf{w}).$$

$$\diamond \mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}.$$

$$\diamond (\mathbf{u} \times \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{v} \cdot \mathbf{w})\mathbf{u}.$$

$$\diamond \mathbf{u} \times (\mathbf{v} \times \mathbf{w}) + \mathbf{v} \times (\mathbf{w} \times \mathbf{u}) + \mathbf{w} \times (\mathbf{u} \times \mathbf{v}) = \mathbf{0}.$$

Proof:

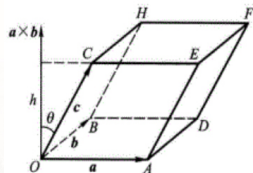
15



Scalar Triple Product of Vectors

Definition. The **scalar triple product** of three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$ is defined to be $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$.

Remark: The scalar triple product of vectors is a scalar!



Proposition. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be vectors in $\mathbb{R}^3$.

- ◇ $|(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}|$ is the volume of the parallelepiped determined by $\mathbf{a}, \mathbf{b}, \mathbf{c}$.
- ◇ If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ satisfy the “right-hand rule”, then $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} > 0$;
If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ satisfy the “left-hand rule”, then $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} < 0$;
- ◇ If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are coplanar vectors, then $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$.

Remark: $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a} = (\mathbf{c} \times \mathbf{a}) \cdot \mathbf{b}$.



Scalar Triple Product of Vectors

- Definition: Let $A = \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}$. Then its **determinant** $\begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$, or **$\det(A)$** , is defined to be the “signed volume” of the parallelepiped spanned by its row (or column) vectors.

Theorem. Suppose that

$$\mathbf{u} = (u_1, u_2, u_3), \mathbf{v} = (v_1, v_2, v_3), \mathbf{w} = (w_1, w_2, w_3).$$

Then the scalar triple product

$$\begin{aligned} \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) &= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \\ &= u_1 \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix} - u_2 \begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix} + u_3 \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} \\ &= u_1 v_2 w_3 + u_2 v_3 w_1 + u_3 v_1 w_2 - u_3 v_2 w_1 - u_2 v_1 w_3 - u_1 v_3 w_2. \end{aligned}$$

Proof:

16



Technique for 2×2 and 3×3 Determinants

- A useful technique:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Attention: Not correct for determinants of higher order!

Example. Evaluate

$$\begin{vmatrix} 1 & 0 & 3 \\ 2 & 5 & 1 \\ 0 & -1 & 2 \end{vmatrix}.$$

Solution:

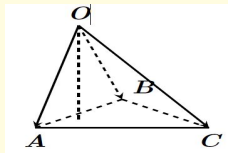
17



An Application

Example. Write $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = \mathbf{c}$. Find the distance from O to the plane where A, B, C locate.

Solution:



18

Example. Let $\mathbf{u} = (\mathbf{i} - 2\mathbf{k}) \times (\mathbf{i} + 3\mathbf{j} - 4\mathbf{k})$ and $\mathbf{v} = 3\mathbf{i} - 12\mathbf{j} + 4\mathbf{k}$. Evaluate the norm of $\text{proj}_{\mathbf{v}}(\mathbf{u})$.

Solution:

19



Chapter 2. Euclidean Vector Spaces

§2.1 Basics of Vectors

§2.2 Dot Product and Orthogonality

§2.3 Cross Product and Scalar Triple Product

§2.4 General Determinants

§2.5 Properties of Determinants; Cramer's Rule

§2.6 Matrix Transformations



Cofactors for a Matrix

Definition. Let $A = [a_{ij}]$ be a square matrix.

- ◇ The **minor** of entry a_{ij} , denoted by M_{ij} , is defined to be the **determinant** of the submatrix that remains after deleting i th row and j th column from A .
- ◇ The **cofactor** of the entry a_{ij} , denoted by C_{ij} , is the number $(-1)^{i+j} M_{ij}$.

$$M_{ij} = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{in} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nj} & \dots & a_{nn} \end{bmatrix}$$

j-th column

i-th row

$$\begin{bmatrix} + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Example. Find the cofactor of a_{11} , a_{22} , a_{31} , and a_{32} of the matrix

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 5 & 8 \\ -2 & 6 & 9 \end{bmatrix}.$$

Solution:

20



Definition of a General Determinant

- For 1×1 matrix, we define $\det([a]) = a$.

Definition. Let $A \in M_n$. For any given i_0 or given j_0 , the **determinant** of A is defined (by induction) to be

$$\begin{aligned}\det(A) &= \sum_{j=1}^n (-1)^{i_0+j} a_{i_0j} M_{i_0j} = \sum_{j=1}^n a_{i_0j} C_{i_0j} \\ &= \sum_{i=1}^n (-1)^{i+j_0} a_{ij_0} M_{ij_0} = \sum_{i=1}^n a_{ij_0} C_{ij_0}.\end{aligned}$$

The right-hand sides are called **cofactor expansions** of A along the i_0 th row or j_0 th column, respectively.

Remark: The result does not depend on the row (or column) that is chosen.

Example. Find the determinant of A by cofactor expansion along the 2nd row or the 3rd column, respectively. Here

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}$$

Solution:

21



Examples of Determinants

Example. Find the following determinant by expanding $\det(A)$ smartly.

$$\begin{vmatrix} 1 & 0 & 0 & -1 \\ 3 & 1 & 2 & 2 \\ 1 & 0 & -2 & 1 \\ 2 & 0 & 0 & 1 \end{vmatrix}.$$

Solution:

22


$$D_n = \begin{vmatrix} a_n & & & & b_{n-1} & & b_n \\ & a_{n-1} & & & & & \\ & & \ddots & & & & \\ & & & a_1 & b_1 & & \\ & & & c_1 & d_1 & & \\ & & & & & \ddots & \\ & & & & & & d_{n-1} \\ c_n & c_{n-1} & & & & & d_n \end{vmatrix}$$

23



Determinants of Triangular Matrices

Example. Find the determinant of A , where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

Solution:

24

Theorem. The determinant of a triangular matrix is the product of the entries on the diagonal line.



Some Basic Theorems

Theorem. If a square matrix A has a row of zeros, or a column of zeros, then $\det(A) = 0$.

Theorem. If a square matrix A has two identical rows or columns, then $\det(A) = 0$.

Idea:

②5

Theorem. For a square matrix A , it satisfies that

$$\det(A) = \det(A^T).$$

Idea:

②6

Problem*: Try to write out the detailed proof by using induction.

Remark: We can proceed with either row operations or column operations.



Elementary Row Operations

Theorem. Let A be an $n \times n$ matrix.

- ◇ If B is the matrix that results when a single row of A is multiplied by a scalar c , then $\det(B) = c \cdot \det(A)$.
- ◇ If B is the matrix that results when two rows of A are interchanged, then $\det(B) = -\det(A)$.
- ◇ If B is the matrix that results when a multiple of one row of A is added to another row, then $\det(B) = \det(A)$.

Explanation:

(27)

Problem*: Try to write out the detailed proof of the second statement by using induction.

Remark: The same results hold for elementary column operations.



Determinants of Elementary Matrices

Corollary. For elementary matrices of order n ,

$$\det(F_i(c)) = c \ (c \neq 0), \quad \det(F_{i,j}) = -1, \quad \det(F_{i,j}(c)) = 1.$$

Moreover, for any matrix $A \in M_n$,

$$\det(F_i(c)A) = \det(F_i(c)) \det(A) \ (c \neq 0),$$

$$\det(F_{i,j}A) = \det(F_{i,j}) \det(A),$$

$$\det(F_{i,j}(c)A) = \det(F_{i,j}(c)) \det(A).$$

Example.

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 3, \quad \begin{vmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix} = -1, \quad \begin{vmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1$$



More Examples

Example. Evaluate

$$\begin{vmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{vmatrix}, \quad \begin{vmatrix} 1 & 0 & 0 & 3 \\ 2 & 7 & 0 & 6 \\ 0 & 6 & 3 & 0 \\ 7 & 3 & 1 & -2 \end{vmatrix}.$$

Solution:

28



More Examples

Example. Evaluate Vandermonde determinant

$$\Delta_n = \begin{vmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \\ x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & \vdots & \vdots \\ x_1^{n-1} & x_2^{n-1} & \dots & x_n^{n-1} \end{vmatrix}$$

Solution:

29

Problem*: Express the following determinant in terms of $\cos \alpha$, $\cos \beta$ and $\cos \gamma$:

$$\begin{vmatrix} 1 & \cos \alpha & \cos 2\alpha \\ 1 & \cos \beta & \cos 2\beta \\ 1 & \cos \gamma & \cos 2\gamma \end{vmatrix}.$$



More Problems

Problem*: Evaluate the following determinants of order n .

$$A_n = \begin{vmatrix} a_1 & 1 & 1 & \dots & 1 \\ 1 & a_2 & & & \\ 1 & & a_3 & & \\ \vdots & & & \ddots & \\ 1 & & & & a_n \end{vmatrix}, \quad B_n = \begin{vmatrix} a_1 & b & b & \dots & b \\ c & a_2 & b & \dots & b \\ c & c & a_3 & \dots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c & c & c & \dots & a_n \end{vmatrix}.$$

$$C_n = \begin{vmatrix} a & b & & & \\ c & a & b & & \\ & c & \ddots & \ddots & \\ & & \ddots & \ddots & b \\ & & & c & a \end{vmatrix}, \quad D_n = \begin{vmatrix} 1+a_1 & a_2 & \dots & a_n \\ a_1 & 1+a_2 & \dots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \dots & 1+a_n \end{vmatrix}$$



Chapter 2. Euclidean Vector Spaces

§2.1 Basics of Vectors

§2.2 Dot Product and Orthogonality

§2.3 Cross Product and Scalar Triple Product

§2.4 General Determinants

§2.5 Properties of Determinants; Cramer's Rule

§2.6 Matrix Transformations



Determinant of a Matrix Product

Researching direction: What about $\det(cA)$, $\det(A + B)$, or $\det(AB)$?

Example. Let $A \in M_n$ and c be a scalar. Then $\det(cA) = c^n \det(A)$.

Theorem. Let $A, B \in M_n$. Then $\det(AB) = \det(A) \det(B)$.

Proof:

③

Corollary. A square matrix A is invertible if and only if $\det(A) \neq 0$.

Corollary. If A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

Problem*: Suppose that $\det(A) = 3$, $\det(B) = 2$ and $\det(A^{-1} + B) = 2$.
Evaluate $\det(A + B^{-1})$.



Adjunct of a Matrix

- Aim: To find an abstract expression for A^{-1} when A is invertible.

Example. Let $A = [a_{ij}] \in M_3$. Prove that

$$a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = \det(A);$$

$$a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23} = 0.$$

$$a_{11}C_{31} + a_{12}C_{32} + a_{13}C_{33} = 0.$$

Proof:

31

Conclusion:

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \end{bmatrix} = \det(A) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} = \det(A) \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} = \det(A) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Adjunct of a Matrix

Definition. If A is any $n \times n$ matrix and C_{ij} is the cofactor of a_{ij} , then the matrix

$$\begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

is called the **adjunct** of A , and is denoted by **adj**(A).

Remark: Pay attention to the **subscripts**!

Example. Find $\text{adj}(A)$ by definition, where $A = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix}$.

Solution:

32



Inverse in Terms of an Adjunct

Theorem. Let $A \in M_n$. Then

$$A \operatorname{adj}(A) = \operatorname{adj}(A) A = \det(A) I_n.$$

Remark: If A is invertible, then $A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$.

Example. Suppose that the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible.

Find A^{-1} by using its adjunct.

Solution:

33



Inverse in Terms of an Adjunct

Example. Find A^{-1} using the adjunct of A , where

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix}.$$

Solution:

34

Problem*: Let $A = \begin{bmatrix} 2 & 1 & -1 \\ -1 & 2 & 1 \\ 1 & -1 & 2 \end{bmatrix}.$

Suppose that $3\text{adj}(A)B = 7A^{-1} + 14B$. Find B .



Determinant of an Adjunct

Question: What is the relation between $\det(A)$ and $\det(\text{adj}(A))$?

Example. Let A be a 4×4 matrix such that

$$\text{adj}(A) = \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ 4 & 0 & 3 & 2 \\ -2 & 0 & -1 & 2 \end{bmatrix}.$$

Find (1) $\det(A)$; (2) A .

Solution:

35

$$\det(\text{adj}(A)) = (\det(A))^{n-1}.$$

Problem*: Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Suppose that $B \in M_3$ satisfies

$$A\text{B}\text{adj}(A) = 2B\text{adj}(A) + I_3.$$

Evaluate $\det(B)$.



Recall: Inverses of Square Matrix of Order 2

Example. Suppose that $ad - bc \neq 0$. Solve the following linear system

$$\begin{cases} ax + by = u \\ cx + dy = v. \end{cases}$$

Solution:

23



Cramer's Rule

Theorem. (Cramer's Rule) If $A\mathbf{x} = \mathbf{b}$ is a linear system with A an invertible square matrix of order n , then the system has a unique solution given by

$$x_i = \frac{\det(A_i)}{\det(A)},$$

for $i = 1, \dots, n$. Here A_i is the matrix obtained by replacing the i th column of A by $\mathbf{b}$.

Proof:

36



An Example

Example. Using Cramer's Rule to solve the following linear system:

$$\begin{aligned}x_1 &+ 2x_3 = 6 \\-3x_1 + 4x_2 + 6x_3 &= 30 \\-x_1 - 2x_2 + 3x_3 &= 8\end{aligned}$$

Solution:

37

Remark: On computers, solving by Cramer's rule is usually less efficient than solving by Gauss elimination.



Equivalence Theorem (Continued)

Theorem. If A is an $n \times n$ matrix, then the following statements are equivalent.

- (a) A is invertible.
- (b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (c) The reduced row echelon form of A is I_n .
- (d) A can be expressed as a product of elementary matrices.
- (e) $A\mathbf{x} = \mathbf{b}$ is consistent for every $n \times 1$ matrix $\mathbf{b}$.
- (f) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $n \times 1$ matrix $\mathbf{b}$.
- (g) $\det(A) \neq 0$.



An Application

Corollary. Let $(a_1, b_1), \dots, (a_n, b_n)$ be n points on the xy -plane. Suppose that $a_1, a_2, \dots, a_n$ are distinct, and $b_1, b_2, \dots, b_n$ are not all zero. Then there is a unique non-zero polynomial $p(x)$ of degree $\leq n - 1$ such that

$$p(a_i) = b_i, \quad (1 \leq i \leq n).$$

Proof:

38

Problem: Write down this polynomial explicitly by (1) Cramer's Rule; (2) Lagrange Interpolation.



Split Sums in a Column

Theorem.

$$\begin{vmatrix} a_{11} & \dots & \hat{a}_{1k} + \tilde{a}_{1k} & \dots & a_{1n} \\ a_{21} & \dots & \hat{a}_{1k} + \tilde{a}_{1k} & \dots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \dots & \hat{a}_{nk} + \tilde{a}_{nk} & \dots & a_{nn} \end{vmatrix} \\ = \begin{vmatrix} a_{11} & \dots & \hat{a}_{1k} & \dots & a_{1n} \\ a_{21} & \dots & \hat{a}_{1k} & \dots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \dots & \hat{a}_{nk} & \dots & a_{nn} \end{vmatrix} + \begin{vmatrix} a_{11} & \dots & \tilde{a}_{1k} & \dots & a_{1n} \\ a_{21} & \dots & \tilde{a}_{1k} & \dots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \dots & \tilde{a}_{nk} & \dots & a_{nn} \end{vmatrix}.$$

Proof:

39

Remark: Same rule applies to rows.



An Example

Example. Evaluate $\det(A_n)$, where

$$A_n = \begin{bmatrix} a & b & b & \dots & b \\ c & a & b & \dots & b \\ c & c & a & \dots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c & c & c & \dots & a \end{bmatrix}.$$

Solution:

40

Remark: Try to use as many methods as possible.



Chapter 2. Euclidean Vector Spaces

§2.1 Basics of Vectors

§2.2 Dot Product and Orthogonality

§2.3 Cross Product and Scalar Triple Product

§2.4 General Determinants

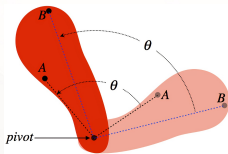
§2.5 Properties of Determinants; Cramer's Rule

§2.6 Matrix Transformations



Recall: Rigid Motion on a Plane

- Core of the Theory of Linear Algebra: To Describe Linear Transforms.



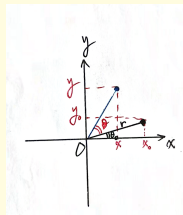
Rigid Rotation



Face Detection/Rotating Images

Example. A mass point has coordinates (x_0, y_0) in the xy -plane. Let us rotate it θ radians counterclockwise around the origin. What is its new location (x, y) ?

Solution:



41



Transformations

Definition. Let V and W be vector spaces. If f is a map (function) with domain V and codomain W , then we say that f is a transformation from V to W and write

$$f : V \rightarrow W.$$

When $W = V$, we also call it an **operator** on V .

Example. A transform $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ always has the coordinate form

$$T(x_1, x_2, \dots, x_n) = (w_1, w_2, \dots, w_m),$$

where

$$\begin{cases} w_1 = f_1(x_1, x_2, \dots, x_n), \\ w_2 = f_2(x_1, x_2, \dots, x_n), \\ \dots \\ w_m = f_m(x_1, x_2, \dots, x_n). \end{cases}$$



Matrix Transformations

A special case: the f_i 's are all linear, i.e.,

$$\begin{cases} w_1 = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n, \\ w_2 = a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n, \\ \dots \\ w_m = a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n. \end{cases}$$

We can write $\mathbf{w} = \mathbf{A}\mathbf{x}$, i.e.,

$$\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Definition. We call the above transformation a **matrix transformation**. Denote (view $\mathbf{x}$ as a column matrix)

$$T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m, \mathbf{x} \mapsto \mathbf{A}\mathbf{x}.$$

Then T_A is called **multiplication by $\mathbf{A}$** , and the matrix is called the **standard matrix** for this transformation.

Notation: We use $[T]$ to denote the standard matrix for a matrix transformation T , i.e., $T(\mathbf{x}) = [T]\mathbf{x}$.



An Example

Example Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by

$$T(x_1, x_2, x_3) = (2x_1 - 3x_2 + x_3, 4x_1 + x_2 - 2x_3).$$

Find the standard matrix $[T]$ for T . And evaluate $T(1, -3, 0)$.

Solution:

42



Standard Matrices for Reflections

- Observation: For the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ for $\mathbb{R}^n$,

$$A = [\begin{array}{c|c|c|c} T_A(\mathbf{e}_1) & T_A(\mathbf{e}_2) & \dots & T_A(\mathbf{e}_n) \end{array}].$$

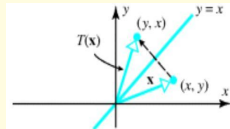
In the following examples, please

- find the standard matrix for each transformation;
- draw the image of the unit square after transformation;
- compare its area with the determinant.

Example.

- (1) Reflection about the y -axis on $\mathbb{R}^2$, i.e., $T(x, y) = (-x, y)$.
- (2) Reflection about the line $y = x$ on $\mathbb{R}^2$, i.e., $T(x, y) = (y, x)$.
- (3) Reflection about the xy -plane on $\mathbb{R}^3$, i.e., $T(x, y, z) = (x, y, -z)$.

Solution:



43

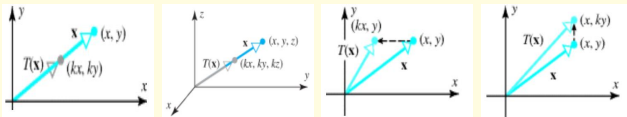


Standard Matrices for Dilations/Expansions

Example.

- (1) Contraction with factor k ($0 < k < 1$) on $\mathbb{R}^2$, i.e., $T(x, y) = (kx, ky)$.
- (2) Dilation with factor k ($k \geq 1$) on $\mathbb{R}^3$, i.e., $T(x, y, z) = (kx, ky, kz)$.
- (3) Compression of $\mathbb{R}^2$ in the x -direction with factor k ($0 \leq k < 1$), i.e.,
$$T(x, y) = (kx, y).$$
- (4) Expansion of $\mathbb{R}^2$ in the y -direction with factor k ($k > 1$), i.e.,
$$T(x, y) = (x, ky).$$

Solution:



44

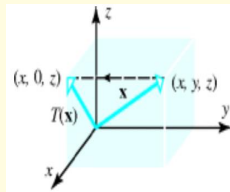


Standard Matrices for Projections

Example.

- (1) Orthogonal projection on the y -axis in $\mathbb{R}^2$, i.e., $T(x, y) = (0, y)$.
- (2) Orthogonal projection on the xz -plane in $\mathbb{R}^3$, i.e., $T(x, y, z) = (x, 0, z)$.

Solution:



45

Problem*: Figure out the orthogonal projection on the plane

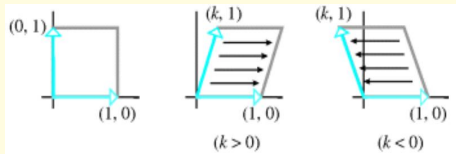
$\Pi : Ax + By + Cz = 0$ in $\mathbb{R}^3$.



Standard Matrices for Shears

Example. The shear of $\mathbb{R}^2$ in the x -direction with factor k , i.e.,
$$T(x, y) = (x + ky, y).$$

Solution:



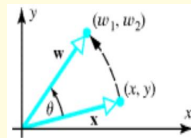
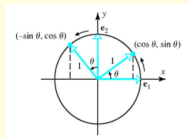
46



Standard Matrices for Rotations

Example. The rotation operator R_θ on $\mathbb{R}^2$ that moves points counterclockwise about the origin through an angle θ .

Solution:



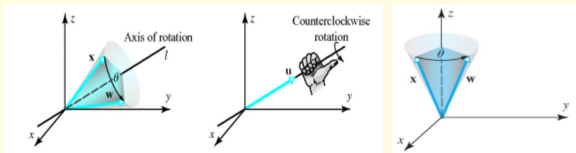
47



Standard Matrices for Rotations

Example. The counterclockwise rotation operator on $\mathbb{R}^3$ about the positive z-axis through an angle θ .

Solution:



48

Fact: Any rotation of vectors in $\mathbb{R}^3$ has at least one axis that does not move. (Try to prove it after learning “eigenvector”.)

Remark: A rotation can be determined by an axis together with an angle.



Properties of Matrix Transformations

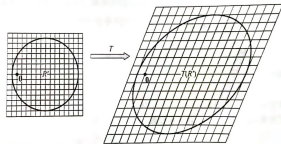
Theorem. For every matrix $A \in M_{m \times n}$, the matrix transformation $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has the following property:

$$T_A(\lambda \mathbf{u} + \mu \mathbf{v}) = \lambda T_A(\mathbf{u}) + \mu T_A(\mathbf{v}), \quad (\forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n, \forall \text{ scalars } \lambda, \mu).$$

Theorem. Let $A \in M_2$ and $T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the corresponding matrix transformation. Suppose that T_A turns a parallelogram region with signed area R into another parallelogram region with signed area R' . Then $R' = \det(A)R$.

Remark: If the area is 0, then it means that the parallelogram degenerates to a line segment or a point.

Remark: The same result holds for the area of a general region, which can be proved by using integration.





Composition of Transformations

Definition. Let U, V, W be three sets and f, g be maps such that

$$U \xrightarrow{f} V \xrightarrow{g} W.$$

Then the **composition** of g with f , denoted by $g \circ f$, is the map $g \circ f : U \rightarrow W$ defined by

$$(g \circ f)(\mathbf{u}) = g(f(\mathbf{u})), \quad (\forall \mathbf{u} \in U).$$

- Fact: Let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^k$, $T_B : \mathbb{R}^k \rightarrow \mathbb{R}^m$ be the multiplication for A, B , respectively. Then

$$T_B \circ T_A = T_{BA}.$$

Moreover, for matrix transformations $T_1, T_2, \dots, T_r$ whose composition exists,

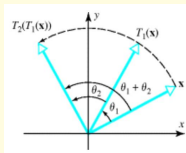
$$[T_r \circ \dots \circ T_2 \circ T_1] = [T_r] \dots [T_2][T_1].$$



Examples of Compositions

Example. Let T_i ($i = 1, 2$) be the rotation operator R_{θ_i} on $\mathbb{R}^2$, respectively. Find the standard matrices for $T_1 \circ T_2$ and $T_2 \circ T_1$.

Solution:



49

Example. On $\mathbb{R}^2$, let T_1 be the reflection about the line $y = x$, and let T_2 be the orthogonal projection on the y -axis. Find the standard matrices for $T_1 \circ T_2$ and $T_2 \circ T_1$.

Solution:

50

Remark: Composition is **NOT** commutative!

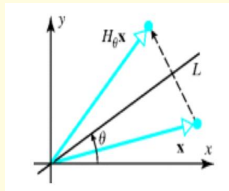
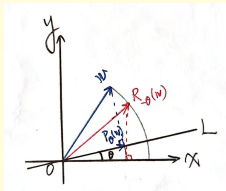


Operators involving an Arbitrary Line

Example. Let L be the line through the origin that makes an angle θ with the positive x -axis. Find the standard matrices for the following operators on $\mathbb{R}^2$.

- (1) P_θ : map each point into its orthogonal projection on L ;
- (2) H_θ : map each point into its reflection about L .

Solution:



51

Remark: Try A. Figure out the whole $T(x, y)$ first, and then find the matrix.
B. Find the matrix first, and then get $T(x, y)$.

Methods: I. Compute directly; II. $P_\theta = R_\theta \circ P_0 \circ R_{-\theta}$; $I + H_\theta = 2P_\theta$.



Injective and Surjective Maps

Definition. Let $f : V \rightarrow W$ be a map from a set V to a set W .

- ◇ We say that f is **injective (one-to-one)**,
if it maps distinct elements in V into distinct elements in W .
- ◇ We say that f is **surjective (onto)**,
if every element in W is in the range of f .
- ◇ We say that f is **bijective**, if it is both injective and surjective.

Remark: $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is injective if and only if:

For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the statement $T_A(\mathbf{x}) = T_A(\mathbf{y})$ implies $\mathbf{x} = \mathbf{y}$.

$T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is surjective if and only if:

For any $\mathbf{b} \in \mathbb{R}^m$, there is some $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{b}$.

Example. Injective/surjective/bijective or not?

- (1) Rotation operators on $\mathbb{R}^2$.
- (2) Dilation with factor k ($k \geq 1$) on $\mathbb{R}^3$.
- (3) Orthogonal projection on the xz -plane in $\mathbb{R}^3$.

Solution:

52



Inverse Map

Definition. If is $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ an bijective matrix operator, then the matrix operator $T_{A^{-1}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that corresponds to A^{-1} is called the **inverse** of T_A . We also denote it by T_A^{-1} , i.e., that $T_A^{-1} = T_{A^{-1}}$.

Remark: For any matrix operator T , we have $T \circ T^{-1} = T^{-1} \circ T = I$.

Example. Find the standard matrix of the inverse of the rotation operator R_θ on $\mathbb{R}^2$.

Solution:

53

Example. Find the inverse of the operator T on $\mathbb{R}^2$ given by

$$T(x, y) = (2x + y, 3x + 4y).$$

Solution:

54



Geometry of Matrix Operators on $\mathbb{R}^2$

Proposition. If E is a 2×2 elementary matrix, then T_E is one of the following:

- (a) A shear along a coordinate axis.
- (b) A reflection about $y = x$.
- (c) A compression along a coordinate axis.
- (d) An expansion along a coordinate axis.
- (e) A reflection about a coordinate axis.
- (f) A compression or expansion along a coordinate axis followed by a reflection about a coordinate axis

Remark: Elementary matrices of order 2 are as follows.

$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}, \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}.$$

Corollary. If T_A is a bijection on $\mathbb{R}^2$, then its geometric effect is the same as an appropriate succession of shears, compressions, expansions, and reflections.